

Generalized Vibrational Analysis

Part I: Linear Algebra on a Metric Space

Goal: establish vectors, covectors, the metric, and projection operators within a general metric space.

Definition 1. Vectors and covectors. Let V be an n -dimensional real vector space with basis $\{e_i\}$. Every vector $v \in V$ has components v^i with $v = v^i e_i$, where repeated upper and lower indices are summed. A *covector* is a linear map $\alpha : V \rightarrow \mathbb{R}$, and the covectors form the *dual space* V^* . The components of a covector are $\alpha_i \equiv \alpha(e_i)$, so that $\alpha(v) = \alpha_i v^i$.

Definition 2. Metric. A *metric* on V is a symmetric, positive-definite bilinear form $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$. Its components are $G_{ij} \equiv \langle e_i, e_j \rangle = G_{ji}$, so that

$$\langle v, w \rangle = G_{ij} v^i w^j.$$

Definition 3. Lowering.¹ The *lowering* (“flat”) of a vector v is the covector $v^\flat \equiv \langle v, \cdot \rangle$, i.e. $v^\flat(w) = \langle v, w \rangle$. The components of a lowered vector are therefore given by $v_i \equiv (v^\flat)_i = G_{ij} v^j$, where the covariant metric G_{ij} acts as the transformation matrix of the lowering operation, $\flat : V \rightarrow V^*$. Note that

$$\langle v, w \rangle = G_{ij} v^i w^j = v_i w^i,$$

i.e. the contraction of one vector with the lowered version of another is equal to their inner product.

Proposition 1. Lowering is invertible.² The lowering operation, $\flat : V \rightarrow V^*$, is a linear and invertible mapping.

Definition 4. Raising. The *raising* (sharp) map $\sharp : V^* \rightarrow V$ is the inverse of $\flat$. Given a vector v which maps to a covector $\alpha \equiv v^\flat$ upon lowering, raising maps the covector image α back to the original vector, $\alpha^\sharp = v$. Since these mappings are inverses of each other, $(v^\flat)^\sharp = v$ and $(\alpha^\sharp)^\flat = \alpha$. The components of a raised covector are given by $\alpha^i \equiv (\alpha^\sharp)^i = G^{ij} \alpha_j$, where the transformation matrix for raising, G^{ij} , must be the inverse of the transformation matrix for lowering, G_{ij} , since $\sharp$ is the inverse of $\flat$:³

$$G^{ik} G_{kj} = G_{jk} G^{ki} = \delta_j^i.$$

As the inverse of a symmetric positive-definite matrix, G^{ij} is also symmetric and positive-definite and thus also a valid metric. This *contravariant metric* serves as a metric for the dual space V^* , where the inner product of covectors can be defined as

$$\langle \beta, \alpha \rangle \equiv G^{ij} \beta_i \alpha_j = \beta_i \alpha^i.$$

Note that the two inner products are consistent with each other: $\langle v, w \rangle = \langle v^\flat, w^\flat \rangle$ and $\langle \beta, \alpha \rangle = \langle \beta^\sharp, \alpha^\sharp \rangle$.⁴

¹*Proof:* By Definition 1 and Definition 2, $v_i = v^\flat(e_i) = \langle v, e_i \rangle = G_{ji} v^j = G_{ij} v^j$ and $\langle v, w \rangle = v^\flat(w) = v_i w^i$.

²*Proof:* Linearity follows from bilinearity of the inner product: given an arbitrary linear combination, $v = c^I v_I$, the inner product satisfies $\langle v, w \rangle = c^I \langle v_I, w \rangle$, which implies $v^\flat = c^I v_I^\flat$. To prove invertibility, first note that *only* the zero vector maps to the zero-covector: $v^\flat = \mathbf{0} \Rightarrow \langle v, v \rangle = v^\flat(v) = 0 \Rightarrow v = \mathbf{0}$. It therefore follows that the mapping is one-to-one: $v^\flat = w^\flat \Rightarrow v^\flat - w^\flat = (v - w)^\flat = \mathbf{0} \Rightarrow v - w = \mathbf{0} \Rightarrow v = w$. This proves the result, because a one-to-one linear mapping between spaces of equal dimension is automatically invertible.

³*Exercise:* Show this by translating $\alpha = v^\flat$ and $v = \alpha^\sharp$ into components and substituting one into the other.

⁴*Exercise:* Verify this.

Projection

Definition 5. Orthogonal projector. An orthogonal projector is a linear map $P : V \rightarrow V$, with components $(Pv)^i = P_j^i v^j$, that is idempotent and self-adjoint:

$$P^2 = P \quad \langle Pv, w \rangle = \langle v, Pw \rangle.$$

Its complement is $Q = 1 - P$.

Theorem 1. Projection onto a vector.⁵ For $v \neq 0$, the orthogonal projector onto $\text{span}\{v\} \subset V$ is given by

$$P_v = \frac{v \otimes v^\flat}{\langle v, v \rangle} \Leftrightarrow (P_v)^i_j = \frac{v^i v_j}{\langle v, v \rangle}$$

and $Q_v = 1 - P_v$ is the orthogonal projector onto the orthogonal complement of v .

Corollary 1. Projection onto a covector.⁶ For $\alpha \neq 0$, the orthogonal projector onto $\text{span}\{\alpha^\sharp\} \subset V$ is given by

$$P_\alpha = \frac{\alpha^\sharp \otimes \alpha}{\langle \alpha, \alpha \rangle} \Leftrightarrow (P_\alpha)^i_j = \frac{\alpha^i \alpha_j}{\langle \alpha, \alpha \rangle}$$

and $Q_\alpha = 1 - P_\alpha$ is the orthogonal projector onto the orthogonal complement of $\alpha^\sharp$.

Definition 6. Projectors acting to the left.⁷ A linear map $P : V \rightarrow V$ also acts to the left on covectors, $\beta P \equiv \beta \circ P$, with components $(\beta P)_j = \beta_i P_j^i$. For an orthogonal projector, self-adjointness means that P commutes with lowering and raising:

$$(Pv)^\flat = v^\flat P \quad (\beta P)^\sharp = P\beta^\sharp \Leftrightarrow G_{ik} P_j^k = P_i^k G_{kj}.$$

Acting to the left, P is therefore an orthogonal projector on V^* , with respect to the covector inner product, onto the lowered image of its range. In particular,

$$\beta P_\alpha = \frac{\langle \beta, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha$$

is the component of β parallel to α , and βQ_α is the component orthogonal to it.

Definition 7. Projection of a bilinear form. The projection of a bilinear form H_{ij} by an orthogonal projector P is $H(Pv, Pw)$, with components

$$(H^P)_{ij} \equiv P_i^k H_{kl} P_j^l.$$

Linear maps between two spaces

From here on, Greek indices μ, ν, ρ, σ refer to a space V of dimension n and Latin indices i, j, k, l refer to a space W of dimension $m \leq n$.

⁵Proof: The equivalence of expressions follows from Definition 3. Note that $v/\langle v, v \rangle = e_v$ is a unit vector in the direction of v . Therefore, $P_v w = \frac{v}{\langle v, v \rangle} \langle v, w \rangle = e_v \langle v, w \rangle = w_\parallel$ is the component of w parallel to v . Therefore, $Q_v w = w - w_\parallel = w_\perp$ is the component of w orthogonal to v . So, these transformations do indeed map onto $\text{span}\{v\}$ and its orthogonal complement, respectively. Idempotency follows from $P_v^2 w = P_v w_\parallel = w_\parallel = P_v w$ and $Q_v^2 w = Q_v w_\perp = w_\perp = Q_v w$. Self-adjointness follows from $\langle P_v w, u \rangle = \frac{\langle v, w \rangle \langle v, u \rangle}{\langle v, v \rangle} = \langle w, P_v u \rangle$.

⁶Proof: This follows by substituting $v = \alpha^\sharp$ into Theorem 1.

⁷Proof: For any w , $(Pv)^\flat(w) = \langle Pv, w \rangle = \langle v, Pw \rangle = v^\flat(Pw) = (v^\flat P)(w)$. Setting $v = \beta^\sharp$ and raising both sides gives the second relation. The rest follows because $\flat$ and $\sharp$ carry idempotency and self-adjointness between V and V^* (Definition 4), and because $\beta P_\alpha = (P_\alpha \beta^\sharp)^\flat$ (Corollary 1).

Definition 8. Linear map and pullback. A linear map $L : V \rightarrow W$ has components $(Lv)^i = L_\mu^i v^\mu$. Its pullback (transpose) $L^* : W^* \rightarrow V^*$ is $L^* \beta \equiv \beta \circ L$, with components $(L^* \beta)_\mu = \beta_i L_\mu^i$.

Proposition 2. Pullback of a surjection is injective.⁸ If L is surjective, then L^* is injective.

Definition 9. Induced metric. Given a metric on V and a surjective $L : V \rightarrow W$, the induced covector inner product on W^* is

$$\langle \beta, \gamma \rangle_W \equiv \langle L^* \beta, L^* \gamma \rangle_V.$$

The induced metric on W is the one whose raising and lowering reproduce this covector inner product.

Proposition 3. Induced metric in components.⁹ The induced contravariant metric

$$G^{ij} = L_\mu^i L_\nu^j G^{\mu\nu}$$

is symmetric and positive-definite. Its inverse G_{ij} is therefore the induced (covariant) metric on W .

Theorem 2. Minimum-norm lift.¹⁰ Let $L : V \rightarrow W$ be surjective and equip W with the induced metric. Define $A : W \rightarrow V$ by $A \equiv \sharp_V \circ L^* \circ \flat_W$, i.e.

$$A_i^\mu = G^{\mu\nu} L_\nu^k G_{ki}.$$

Then: (a) $LA = 1_W$, i.e. $L_\mu^i A_j^\mu = \delta_j^i$; (b) AL is the orthogonal projector on V onto $(\ker L)^\perp$; (c) Aw is the unique vector of minimum norm with $Lv = w$; (d) $\langle Aw, Aw' \rangle_V = \langle w, w' \rangle_W$.

Remark 1. Interpretation of the lift. A is simply L with both of its indices moved by the metrics. When $m = n$, $A = L^{-1}$ and $AL = 1_V$.

⁸Proof: If $L^* \beta = 0$, then $\beta(Lv) = 0$ for all $v \in V$. Since L is surjective, $\beta(w) = 0$ for all $w \in W$, so $\beta = 0$.

⁹Proof: By Definition 8 and Definition 4, $\langle \beta, \gamma \rangle_W = G^{\mu\nu} (\beta_i L_\mu^i) (\gamma_j L_\nu^j) = G^{ij} \beta_i \gamma_j$. Symmetry follows from that of $G^{\mu\nu}$. For $\beta \neq 0$, $L^* \beta \neq 0$ by Proposition 2, so $G^{ij} \beta_i \beta_j = \langle L^* \beta, L^* \beta \rangle_V > 0$. A symmetric positive-definite matrix has a symmetric positive-definite inverse G_{ij} , which is therefore a metric on W . By Definition 4, its raising matrix G^{ij} reproduces the induced covector inner product.

¹⁰Proof: First, for any $v, u \in V$, $\langle ALv, u \rangle_V = G_{\mu\rho} G^{\mu\nu} L_\nu^k G_{ki} L_\sigma^i v^\sigma u^\rho = G_{ki} (Lu)^k (Lv)^i = \langle Lv, Lu \rangle_W$. (†) (a) $L_\mu^i A_j^\mu = L_\mu^i L_\nu^k G^{\mu\nu} G_{kj} = G^{ik} G_{kj} = \delta_j^i$ by Proposition 3. (b) Let $P = AL$. By (a), $P^2 = A(LA)L = P$, and by (†), P is self-adjoint. Also by (a), $LP = L$, so $Pv = 0$ if and only if $Lv = 0$, i.e. $\ker P = \ker L$. The range of a self-adjoint projector is the orthogonal complement of its kernel (if $Pu = 0$, then $\langle Pv, u \rangle = \langle v, Pu \rangle = 0$, and the dimensions add to n). (c) $LAw = w$ by (a). If $Lv = w$, then $Pv = Aw$, so $v = Aw + (1 - P)v$ is an orthogonal decomposition by (b) and $\langle v, v \rangle_V = \langle Aw, Aw \rangle_V + \langle (1 - P)v, (1 - P)v \rangle_V$, which is minimized only by $v = Aw$. (d) By (a) and (†), $\langle Aw, Aw' \rangle_V = \langle ALAw, Aw' \rangle_V = \langle LAw, LAw' \rangle_W = \langle w, w' \rangle_W$.

Part II: Coordinates and Derivatives

Goal: show that coordinate displacements are vectors and derivatives are covectors, so that the machinery of Part I applies to them.

Definition 10. Displacements and differentials. Let q^μ be coordinates on a neighborhood of a reference point in configuration space. Infinitesimal displacements $d\mathbf{q}$, with components dq^μ , are vectors. For a smooth function f , the differential $df(d\mathbf{q}) = \frac{\partial f}{\partial q^\mu} dq^\mu$ is linear in $d\mathbf{q}$, so df is a covector with components $\frac{\partial f}{\partial q^\mu}$.

Definition 11. Coordinate map. A second set of $m \leq n$ coordinates $q^i = q^i(q^\mu)$ has *Jacobian* and *second derivative*

$$B_\mu^i \equiv \frac{\partial q^i}{\partial q^\mu} \quad C_{\mu\nu}^i \equiv \frac{\partial^2 q^i}{\partial q^\mu \partial q^\nu} = C_{\nu\mu}^i.$$

We assume B_μ^i has full row rank at the reference point, so that B is surjective.

Proposition 4. Displacements push forward.¹¹

$$dq^i = B_\mu^i dq^\mu$$

Proposition 5. Gradients pull back.¹² If f depends on q^μ only through q^i , then

$$\frac{\partial f}{\partial q^\mu} = \frac{\partial f}{\partial q^i} B_\mu^i,$$

i.e. df in the q^μ coordinates is the pullback B^* of df in the q^i coordinates.

Remark 2. Transformation laws. When $m = n$ and B is invertible, these become the classical contravariant and covariant transformation laws, and the contraction $\frac{\partial f}{\partial q^i} dq^i = \frac{\partial f}{\partial q^\mu} dq^\mu$ is invariant. When $m < n$, displacements can only be pushed forward and gradients can only be pulled back.

Proposition 6. Second derivatives.¹³ If f depends on q^μ only through q^i , then

$$\frac{\partial^2 f}{\partial q^\mu \partial q^\nu} = \frac{\partial^2 f}{\partial q^i \partial q^j} B_\mu^i B_\nu^j + \frac{\partial f}{\partial q^k} C_{\mu\nu}^k.$$

Second derivatives therefore transform as a bilinear form only when the map is linear ($C = 0$) or at a stationary point ($\partial f / \partial q^k = 0$).

¹¹Proof: This is the chain rule applied to $q^i(q^\mu)$: to first order, $dq^i = \frac{\partial q^i}{\partial q^\mu} dq^\mu$.

¹²Proof: This is the chain rule applied to $f(q^i(q^\mu))$. The components of the pullback are given in Definition 8.

¹³Proof: Differentiate Proposition 5 with respect to q^ν , using the product rule and the chain rule: $\frac{\partial^2 f}{\partial q^i \partial q^j} B_\nu^j B_\mu^i + \frac{\partial f}{\partial q^k} \frac{\partial B_\mu^k}{\partial q^\nu}$, where $\partial B_\mu^k / \partial q^\nu = C_{\mu\nu}^k$.

Part III: The Physical Model

Goal: identify the kinetic energy as the metric, and apply Part I to Cartesian, mass-weighted Cartesian, and internal coordinates.

Definition 12. Cartesian coordinates and kinetic-energy metric. For N atoms with Cartesian coordinates x^μ ($\mu = 1, \dots, 3N$), let $m_{(\mu)}$ denote the mass of the atom to which coordinate μ belongs. Cartesian displacements from a reference geometry are $q^\mu = x^\mu - x_0^\mu$. The *kinetic-energy metric* is

$$G_{\mu\nu} \equiv m_{(\mu)} \delta_{\mu\nu},$$

chosen so that $T = \frac{1}{2} \langle \dot{q}, \dot{q} \rangle$.

Proposition 7. Contravariant Cartesian metric.¹⁴

$$G^{\mu\nu} = m_{(\mu)}^{-1} \delta^{\mu\nu}$$

Definition 13. Mass-weighted coordinates. The mass-weighted Cartesian displacements are $\tilde{q}^\mu \equiv m_{(\mu)}^{\frac{1}{2}} q^\mu$. Mass-weighted quantities are marked with a tilde.

Proposition 8. Mass-weighted metric.¹⁵

$$\tilde{G}_{\mu\nu} = \delta_{\mu\nu} \quad \tilde{G}^{\mu\nu} = \delta^{\mu\nu}$$

Definition 14. Internal coordinates. Internal coordinates q^i ($i = 1, \dots, M$, with $M = 3N - 6$, or $3N - 5$ for linear molecules) are displacements of non-redundant functions of the geometry that are invariant to overall translation and rotation. Their mass-weighted Jacobian (B-matrix) and its derivative (C-tensor) are

$$\tilde{B}_\mu^i \equiv \frac{\partial q^i}{\partial \tilde{q}^\mu} \quad \tilde{C}_{\mu\nu}^i \equiv \frac{\partial^2 q^i}{\partial \tilde{q}^\mu \partial \tilde{q}^\nu},$$

and $\tilde{B}_\mu^i$ has full row rank at the reference geometry. Index-free internal-coordinate objects are marked with a prime, e.g. v' .

Theorem 3. Wilson G-matrix.¹⁶ The metric induced on internal coordinates by $\tilde{B}$ has contravariant components

$$G^{ij} = \delta^{\mu\nu} \tilde{B}_\mu^i \tilde{B}_\nu^j = \sum_\mu m_{(\mu)}^{-1} B_\mu^i B_\mu^j,$$

where $B_\mu^i = \partial q^i / \partial q^\mu$. The covariant internal-coordinate metric G_{ij} is its inverse.

¹⁴*Proof:* By Definition 4, $G^{\mu\nu}$ is the inverse of the diagonal matrix $m_{(\mu)} \delta_{\mu\nu}$.

¹⁵*Proof:* The map $q^\mu \rightarrow \tilde{q}^\mu$ is linear and invertible, with Jacobian $m_{(\mu)}^{\frac{1}{2}} \delta_{\mu\nu}^{\frac{1}{2}}$. For an invertible map, the induced metric is the same inner product expressed in the new components (Theorem 2 (d) with $A = L^{-1}$), so Proposition 3 and Proposition 7 give $\tilde{G}^{\mu\nu} = m_{(\mu)}^{\frac{1}{2}} m_{(\nu)}^{\frac{1}{2}} m_{(\mu)}^{-1} \delta^{\mu\nu} = \delta^{\mu\nu}$, whose inverse is $\tilde{G}_{\mu\nu} = \delta_{\mu\nu}$. As a check, $T = \frac{1}{2} \sum_\mu m_{(\mu)} (\dot{q}^\mu)^2 = \frac{1}{2} \sum_\mu (\dot{\tilde{q}}^\mu)^2$.

¹⁶*Proof:* By Proposition 4, displacements map as $dq^i = \tilde{B}_\mu^i d\tilde{q}^\mu$, and $\tilde{B}$ is surjective (Definition 14). Proposition 3 with $L_\mu^i = \tilde{B}_\mu^i$ and Proposition 8 give the first form. For the second, $q^\nu = m_{(\nu)}^{-\frac{1}{2}} \tilde{q}^\nu$ and Proposition 5 give $\tilde{B}_\mu^i = m_{(\mu)}^{-\frac{1}{2}} B_\mu^i$.

Corollary 2. *A-matrix.*¹⁷ *The A-matrix*

$$\tilde{A}_i^\mu \equiv \delta^{\mu\nu} \tilde{B}_\nu^k G_{ki}$$

satisfies $\tilde{B}_\mu^i \tilde{A}_j^\mu = \delta_j^i$, and $\tilde{A}_i^\mu \tilde{B}_\nu^i$ is the orthogonal projector onto the vibrational subspace $(\ker \tilde{\mathbf{B}})^\perp$, where $\ker \tilde{\mathbf{B}}$ is spanned by infinitesimal translations and rotations.

Proposition 9. *Kinetic energy in internal coordinates.*¹⁸ *For Cartesian velocities orthogonal to $\ker \tilde{\mathbf{B}}$ (no infinitesimal overall translation or rotation, i.e. the linearized Eckart conditions),*

$$T = \frac{1}{2} G_{ij} \dot{q}^i \dot{q}^j.$$

¹⁷*Proof:* Apply Theorem 2 (a) and (b) with $\mathbf{L} = \tilde{\mathbf{B}}$ and $G^{\mu\nu} = \delta^{\mu\nu}$ (Proposition 8). Each q^i is invariant to overall translation and rotation, so its derivative along an infinitesimal translation or rotation vanishes and these lie in $\ker \tilde{\mathbf{B}}$. By rank-nullity, $\dim \ker \tilde{\mathbf{B}} = 3N - M$, which is the number of independent infinitesimal translations and rotations, so they span the kernel.

¹⁸*Proof:* By Proposition 4, $\dot{q}^i = \tilde{B}_\mu^i \dot{\tilde{q}}^\mu$. If $\dot{\tilde{\mathbf{q}}} \perp \ker \tilde{\mathbf{B}}$, then $\dot{\tilde{\mathbf{q}}} = \tilde{\mathbf{A}} \tilde{\mathbf{B}} \dot{\tilde{\mathbf{q}}} = \tilde{\mathbf{A}} \dot{\tilde{\mathbf{q}}}'$ by Corollary 2, and by Theorem 2 (d), $T = \frac{1}{2} \langle \dot{\tilde{\mathbf{q}}}, \dot{\tilde{\mathbf{q}}} \rangle = \frac{1}{2} \langle \tilde{\mathbf{A}} \dot{\tilde{\mathbf{q}}}', \tilde{\mathbf{A}} \dot{\tilde{\mathbf{q}}}' \rangle = \frac{1}{2} G_{ij} \dot{q}^i \dot{q}^j$.

Part IV: Vibrational Analysis

Goal: transform the gradient and Hessian into internal coordinates, set up the GF eigenproblem, and project the gradient direction out of the Hessian.

Definition 15. Gradient and Hessian. The gradient and Hessian in each coordinate system are defined by the Taylor expansion of the potential energy about the reference geometry:

$$V = V_0 + g_\mu q^\mu + \frac{1}{2} H_{\mu\nu} q^\mu q^\nu + \mathcal{O}(q^3)$$

$$V = V_0 + \tilde{g}_\mu \tilde{q}^\mu + \frac{1}{2} \tilde{H}_{\mu\nu} \tilde{q}^\mu \tilde{q}^\nu + \mathcal{O}(\tilde{q}^3)$$

$$V = V_0 + g_i q^i + \frac{1}{2} H_{ij} q^i q^j + \mathcal{O}(q'^3)$$

Proposition 10. Mass-weighted gradient and Hessian.¹⁹

$$\tilde{g}_\mu = m_{(\mu)}^{-\frac{1}{2}} g_\mu \quad \tilde{H}_{\mu\nu} = m_{(\mu)}^{-\frac{1}{2}} m_{(\nu)}^{-\frac{1}{2}} H_{\mu\nu}$$

Theorem 4. Internal-coordinate gradient.²⁰

$$g_i = \tilde{g}_\mu \tilde{A}_i^\mu$$

Theorem 5. Internal-coordinate Hessian.²¹

$$H_{ij} = (\tilde{H}_{\mu\nu} - g_k \tilde{C}_{\mu\nu}^k) \tilde{A}_i^\mu \tilde{A}_j^\nu$$

The correction term arises because the map to internal coordinates is non-linear (Proposition 6).

Definition 16. GF matrix. The GF matrix is the internal-coordinate Hessian with its first index raised,

$$(GF)_j^i \equiv G^{ik} H_{kj},$$

a linear map on internal-coordinate displacements. Its eigenvalues are the squared harmonic frequencies ω^2 .

Proposition 11. GF and mass-weighted Cartesian eigenvalues.²² Let $\tilde{P}_\nu^\mu = \tilde{A}_i^\mu \tilde{B}_\nu^i$. The eigenvalues of $G^{ik} H_{kj}$ are the eigenvalues of $\tilde{P}_\mu^\rho (\tilde{H}_{\rho\sigma} - g_k \tilde{C}_{\rho\sigma}^k) \tilde{P}_\nu^\sigma$ restricted to the vibrational subspace. At a stationary point, these are the eigenvalues of the projected mass-weighted Hessian.

¹⁹Proof: The Taylor coefficients are the derivatives at the reference geometry, e.g. $g_\mu = \partial V / \partial q^\mu$. The map $\tilde{q}^\mu \rightarrow q^\mu = m_{(\mu)}^{-\frac{1}{2}} \tilde{q}^\mu$ is linear, so Proposition 5 and Proposition 6 (with $C = 0$) apply with Jacobian $m_{(\mu)}^{-\frac{1}{2}} \delta_\nu^\mu$.

²⁰Proof: The potential is invariant to overall translation and rotation, so near the reference geometry it depends on $\tilde{q}^\mu$ only through the q^i . By Proposition 5, $\tilde{g}_\mu = g_k \tilde{B}_\mu^k$. Contracting with $\tilde{A}_i^\mu$ and using Corollary 2 gives $\tilde{g}_\mu \tilde{A}_i^\mu = g_k \delta_i^k = g_i$.

²¹Proof: As in Theorem 4, Proposition 6 gives $\tilde{H}_{\mu\nu} - g_k \tilde{C}_{\mu\nu}^k = H_{kl} \tilde{B}_\mu^k \tilde{B}_\nu^l$. Contracting with $\tilde{A}_i^\mu \tilde{A}_j^\nu$ and using Corollary 2 gives the result.

²²Proof: Let $K_{\mu\nu} = \tilde{H}_{\mu\nu} - g_k \tilde{C}_{\mu\nu}^k = H_{kl} \tilde{B}_\mu^k \tilde{B}_\nu^l$ (proof of Theorem 5). Since $\tilde{B}_\rho^k \tilde{P}_\mu^\rho = \tilde{B}_\mu^k$ by Corollary 2, $\tilde{P}_\mu^\rho K_{\rho\sigma} \tilde{P}_\nu^\sigma = K_{\mu\nu}$; the projectors only make explicit that K vanishes on $\ker \tilde{B}$. Raise the first index of K with $\delta^{\mu\nu}$ to get a linear map $\mathbf{K}$ on V . For any internal u^j , $\delta^{\mu\rho} K_{\rho\nu} \tilde{A}_j^\nu u^j = \delta^{\mu\rho} \tilde{B}_\rho^k H_{kj} u^j$ and $\tilde{A}_i^\mu u^i = \delta^{\mu\rho} \tilde{B}_\rho^k G_{ki} u^i$. Since $\tilde{B}^*$ is injective (Proposition 2), $\mathbf{K} \tilde{A} u' = \lambda \tilde{A} u'$ if and only if $H_{kj} u^j = \lambda G_{ki} u^i$, i.e. $G^{ik} H_{kj} u^j = \lambda u^i$. By Theorem 2, $\tilde{A}$ maps internal displacements bijectively onto $(\ker \tilde{B})^\perp$, so the two eigenproblems are equivalent. At a stationary point, $g_k = 0$ and $K_{\mu\nu} = \tilde{H}_{\mu\nu}$.

Definition 17. Gradient-direction projector. For $g_i \neq 0$, with $g^i = G^{ij}g_j$ and $\langle g', g' \rangle = G^{ij}g_i g_j$, the gradient-direction projector and its complement are

$$(P_g)^i_j = \frac{g^i g_j}{\langle g', g' \rangle} \quad Q_g = 1 - P_g.$$

Theorem 6. Projected Hessian.²³ The projected Hessian

$$H_{ij}^Q = (Q_g)^k_i H_{kl} (Q_g)^l_j$$

gives a GF matrix $G^{ik} H_{kj}^Q$ with eigenvector g^i of eigenvalue 0. Every eigenvector v^i with non-zero eigenvalue satisfies $g_i v^i = 0$, and these $M - 1$ eigenvalues are the generalized frequencies ω^2 for motion orthogonal to the gradient.

Remark 3. Comparison with Cartesian projection. To investigate: the relation to Cartesian gradient projection (Miller, Handy, and Adams). Note that $\tilde{A}_i^\mu g^i = \tilde{g}^\mu$, so the gradient directions agree. By Proposition 11, however, the internal-coordinate result carries the curvature term $g_k \tilde{C}_{\mu\nu}^k$, so projected frequencies away from stationary points depend on the choice of coordinates.

²³Proof: Write $Q = Q_g$, so that $Q_j^l = \delta_j^l - g^l g_j / \langle g', g' \rangle$. Then $Q_j^l g^j = g^l - g^l = 0$, so $H_{kj}^Q g^j = 0$ and g^i is an eigenvector of eigenvalue 0. The matrix $G^{ik} H_{kj}^Q$ is self-adjoint with respect to G_{ij} , because $G_{li} G^{ik} H_{kj}^Q = H_{lj}^Q$ is symmetric. It therefore has a G -orthogonal eigenbasis containing g^i , and the remaining $M - 1$ eigenvectors satisfy $G_{ij} g^i v^j = g_j v^j = 0$. Every eigenvector with $\lambda \neq 0$ is of this kind, since $\lambda g_i v^i = g^k H_{kj}^Q v^j = 0$. For such v^i , $Q_j^l v^j = v^l$, so H^Q agrees with H on the hyperplane orthogonal to the gradient.